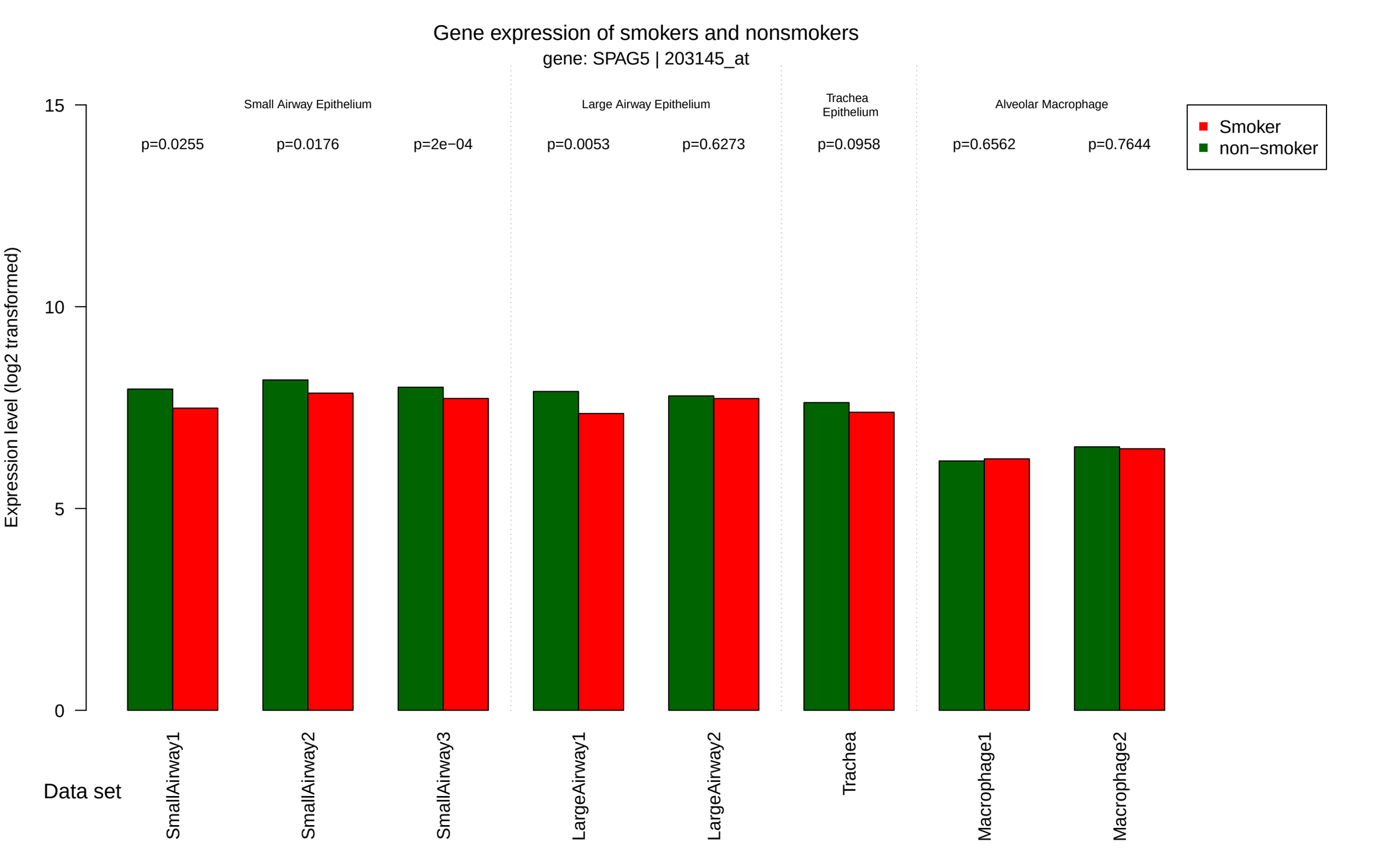
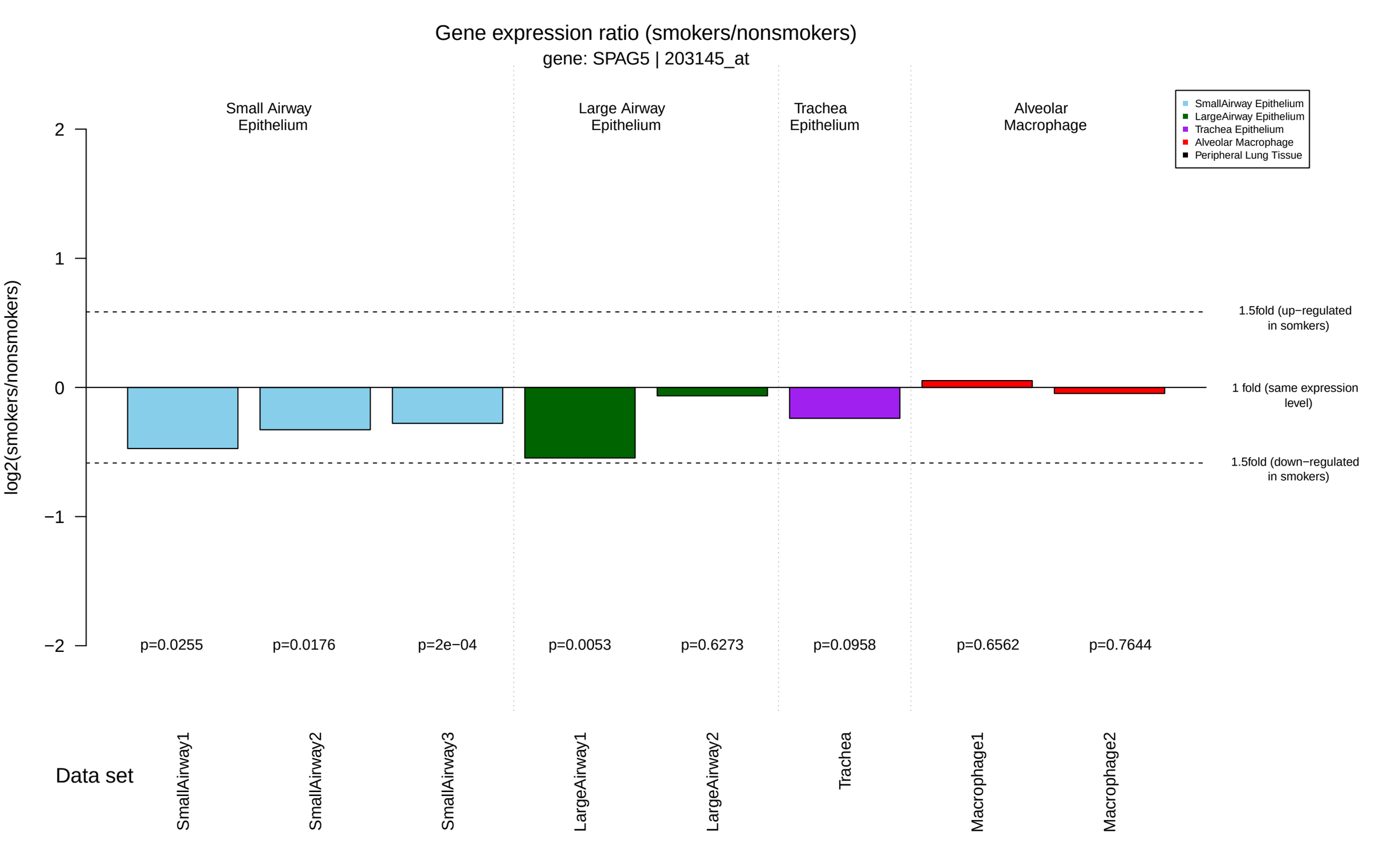
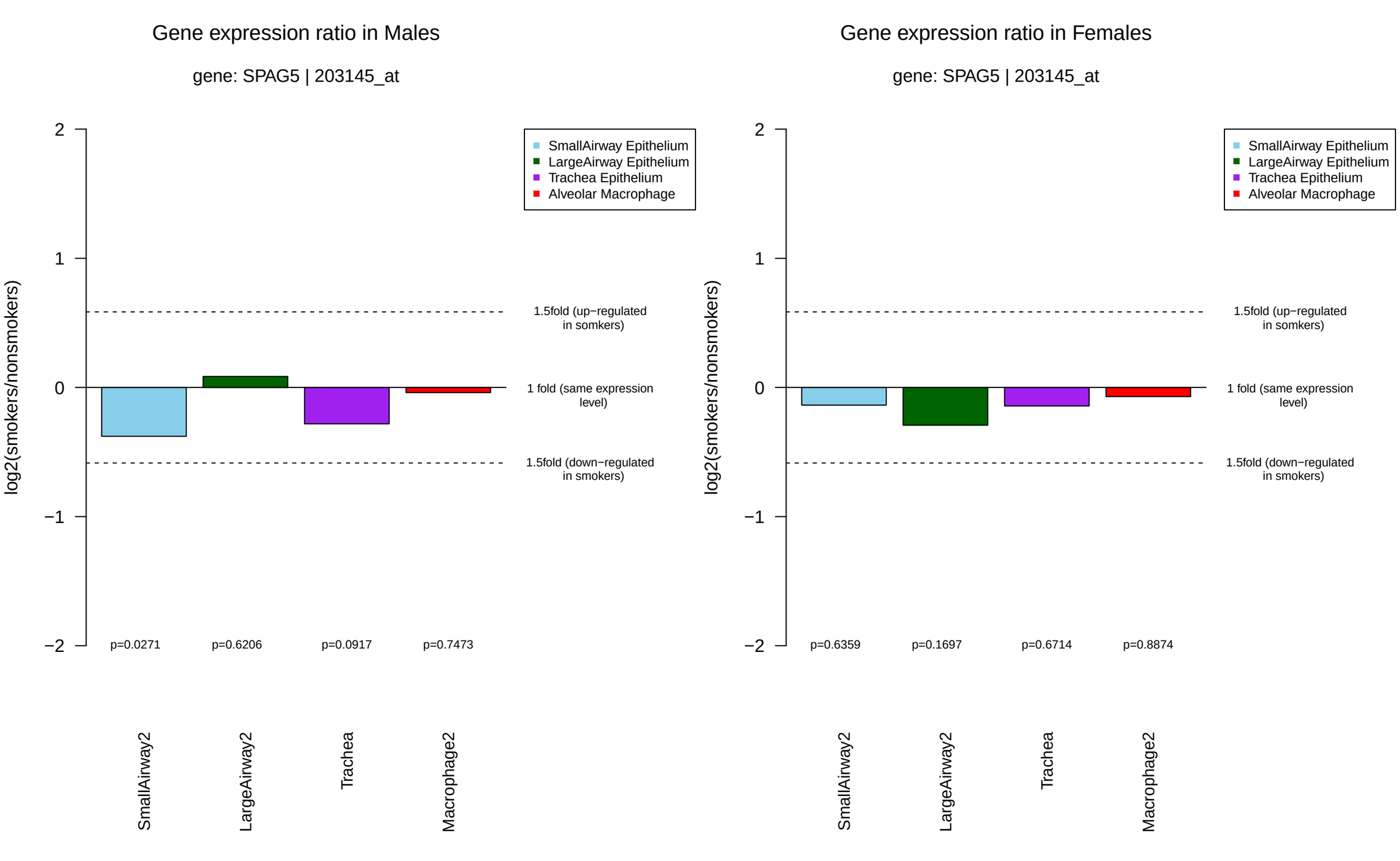
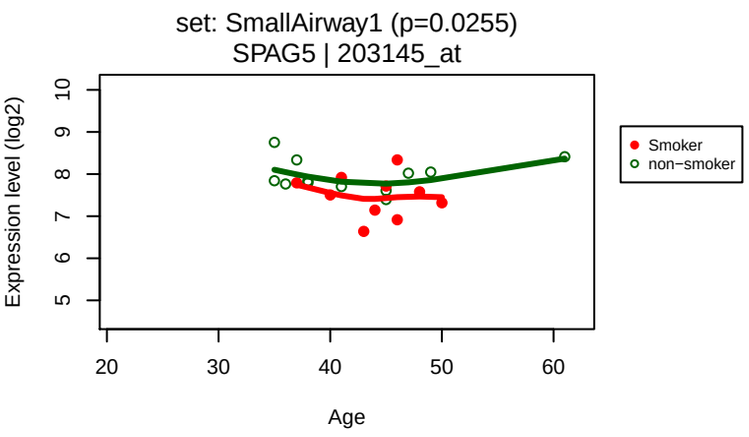
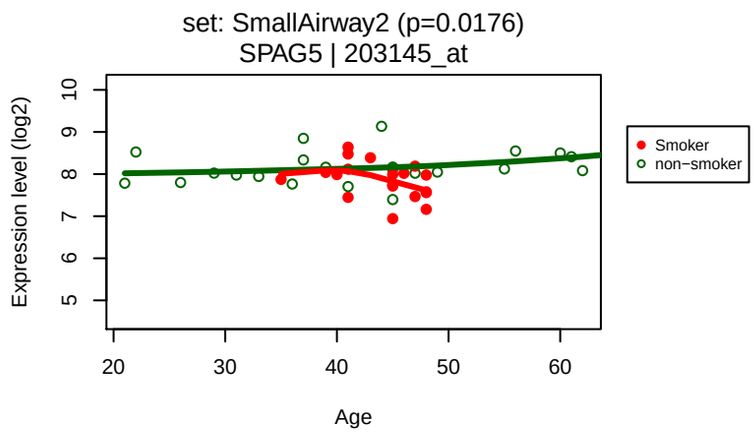
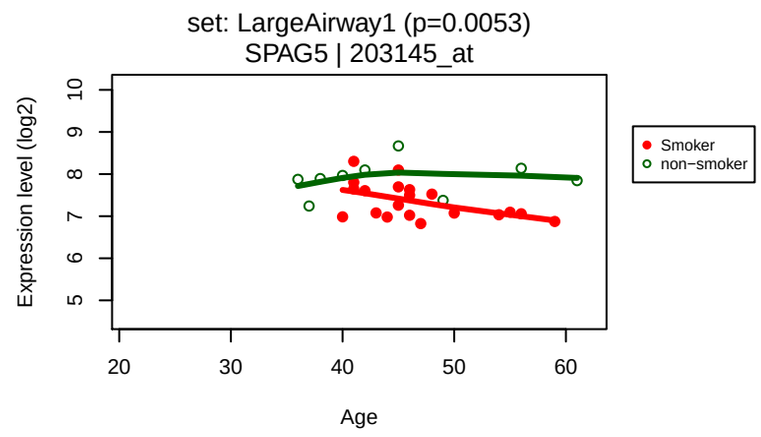

Gene expression of smokers and nonsmokers
gene: SPAG5 | 203145_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0255
p=0.0176
p=2e−04
p=0.0053
p=0.6273
p=0.0958
p=0.6562
p=0.7644
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

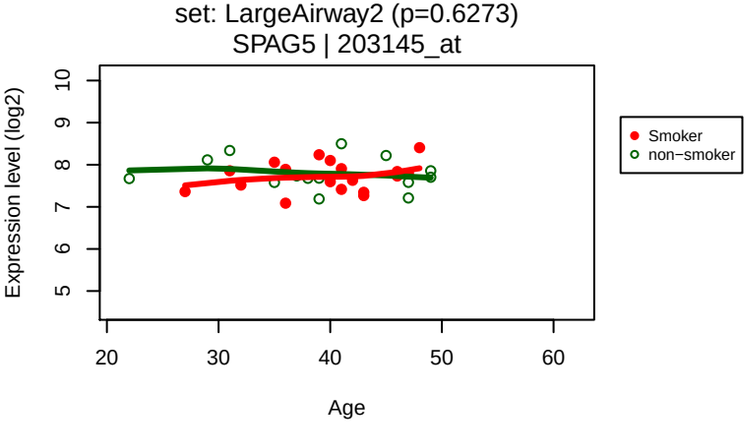

Gene expression ratio (smokers/nonsmokers)
gene: SPAG5 | 203145_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0255
p=0.0176
p=2e−04
p=0.0053
p=0.6273
p=0.0958
p=0.6562
p=0.7644
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

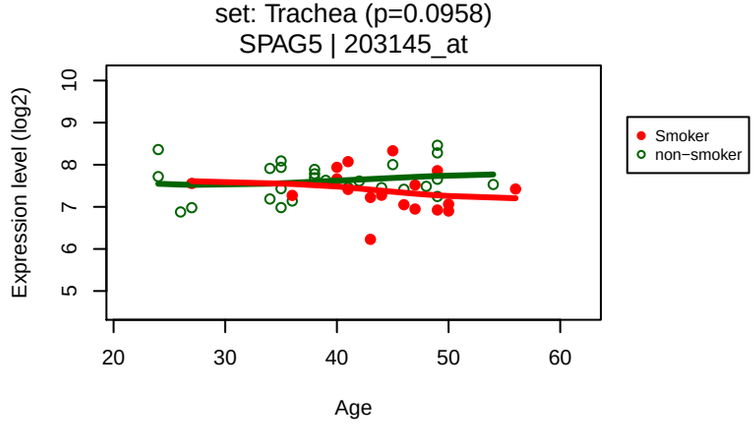

Gene expression ratio in Males
gene: SPAG5 | 203145_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0271
p=0.6206
p=0.0917
p=0.7473
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: SPAG5 | 203145_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6359
p=0.1697
p=0.6714
p=0.8874
SmallAirway2
LargeAirway2
Trachea
Macrophage2

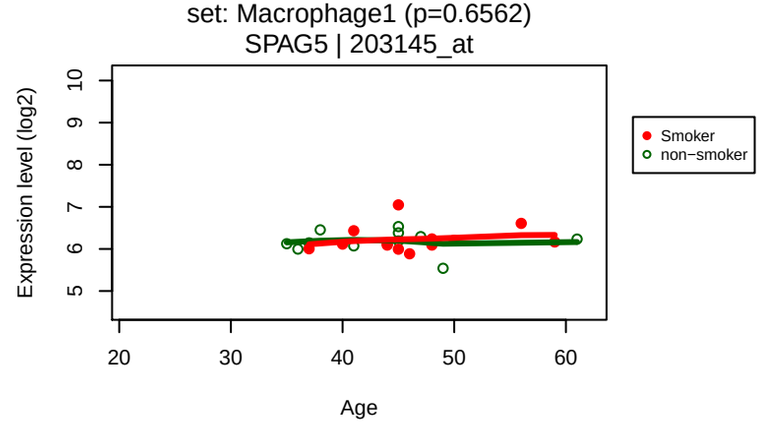

set: SmallAirway1 (p=0.0255)
SPAG5 | 203145_at
Expression level (log2)
Age
Smoker
non−smoker

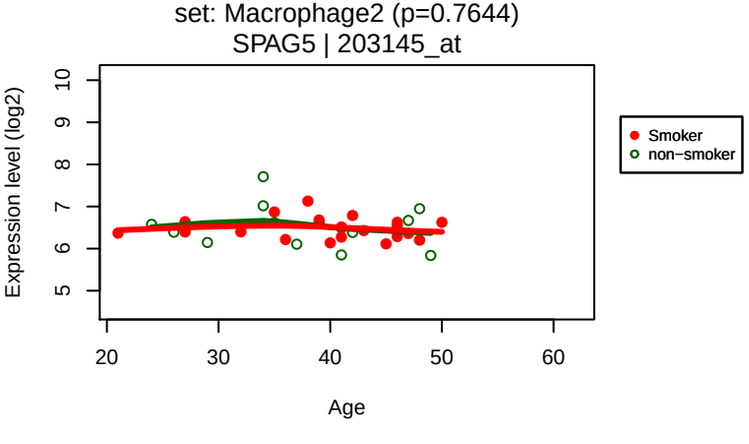

set: SmallAirway2 (p=0.0176)
SPAG5 | 203145_at
Expression level (log2)
Age
Smoker
non−smoker

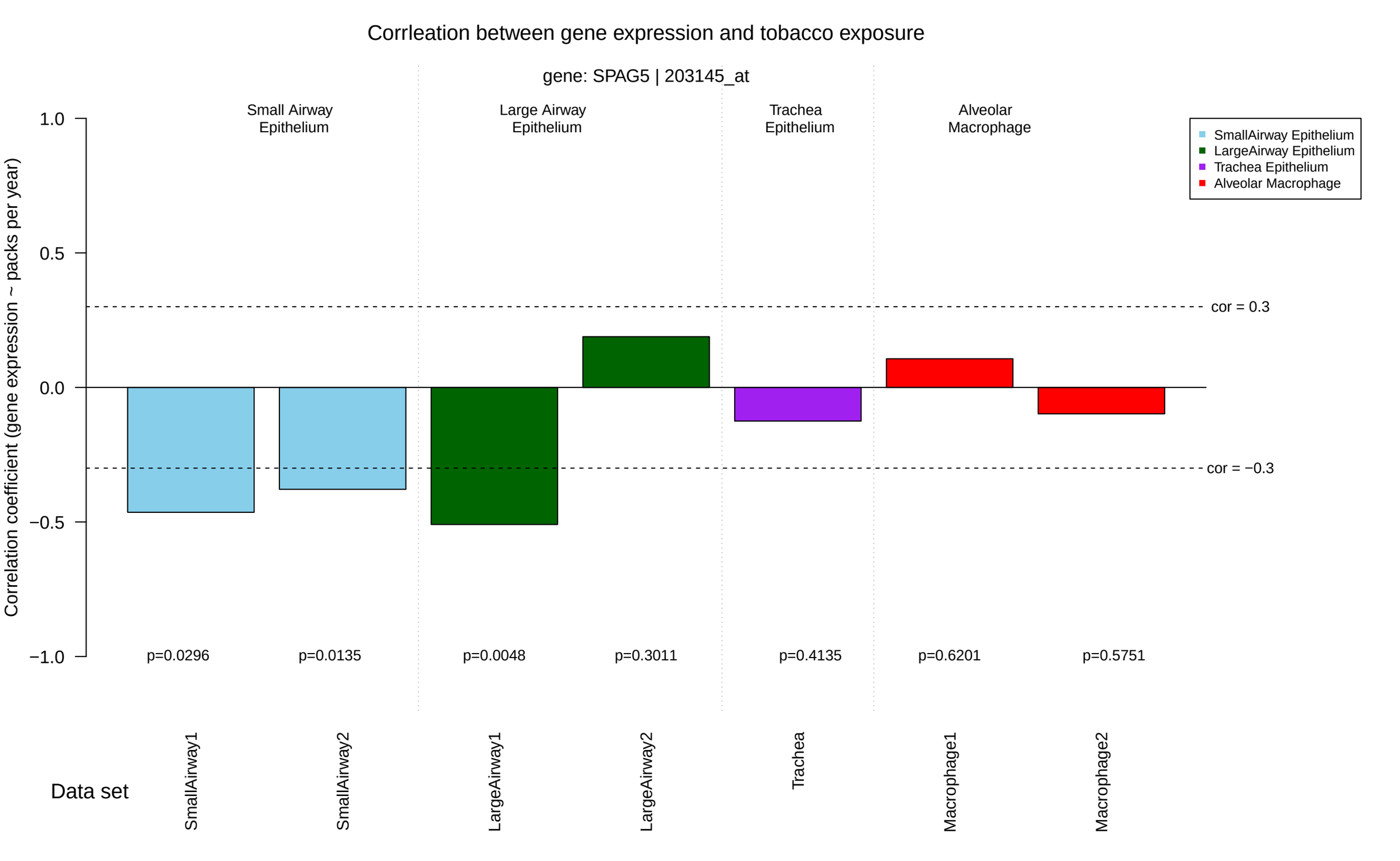

set: LargeAirway1 (p=0.0053)
SPAG5 | 203145_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.6273)
SPAG5 | 203145_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0958)
SPAG5 | 203145_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.6562)
SPAG5 | 203145_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.7644)
SPAG5 | 203145_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: SPAG5 | 203145_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0296
p=0.0135
p=0.0048
p=0.3011
p=0.4135
p=0.6201
p=0.5751
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2